

AYE
TECH HUB

• SOLAR ENERGY • LESSON 07

Charge Controllers

MPPT · PWM · Charging Stages · Sizing

Lesson 07 of 30 | Solar Energy Program

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- MPPT vs PWM: the efficiency difference that can cost you 30% energy
- The 4 charging stages: bulk, absorption, float and equalize
- How MPPT extracts maximum power as temperature and irradiance change
- Step-by-step charge controller sizing for any off-grid system
- Advanced settings: battery type, equalize voltage, temperature compensation

The Charge Controller — Battery Guardian

A charge controller sits between your solar panels and your battery bank. Its two primary jobs: **prevent overcharging** (which destroys batteries) and **manage the charging process** to maximise battery life and energy harvest. In off-grid and hybrid systems, the charge controller is the most critical component for battery longevity — a correctly programmed controller can double the life of a battery bank.



Figure 1 — PWM vs MPPT Charge Controller Comparison

Comparison Point	PWM	MPPT
Efficiency	~75%	93–98%
Panel voltage requirement	Panel Vmp must match battery voltage (12V panels → 12V battery, etc.)	Panel voltage can be much higher than battery (72V string → 24V battery bank — fine)
Energy harvest advantage	None — wastes the voltage difference	Captures 10–30% more energy in cold weather when panel Voc is highest
Cost	Lowest	Higher (2–5×) but pays back in energy gain
Best for	Small systems < 200Wp Panel Vmp close to battery voltage Budget installs	All systems > 200Wp High-voltage panel strings Cold or variable-irradiance climates
Typical brands	Morningstar, Steca, generic Chinese	Victron, Outback, Midnight Solar, EPeve

The MPPT voltage advantage: On a cold morning (5°C), a 400W panel may produce Voc = 50V. A PWM controller locks operation at battery voltage (say 26V) — wasting the extra 24V as heat. An MPPT controller converts that higher voltage into additional current delivered to the battery, capturing the full energy available. This advantage is largest in cold climates and morning/evening hours.

The 4 Charging Stages — How Batteries Are Correctly Charged

Every professional charge controller cycles through four stages. Understanding these stages lets you verify correct controller operation, diagnose problems, and optimise settings for your specific battery chemistry.

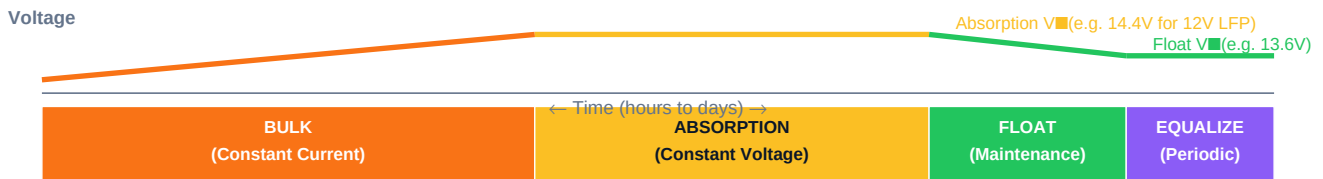


Figure 2 — Battery Charging Stages: Bulk → Absorption → Float → Equalize

Stage	What Happens	When It Ends	Voltage / Current
BULK (Constant Current)	Controller delivers maximum available current from panels. Battery voltage rises steadily. Battery is "empty" — accepts all available current.	Battery reaches absorption voltage setpoint (e.g. 14.4V for 12V LFP)	Current: maximum (Isc of panels) Voltage: rising from low to absorption
ABSORPTION (Constant Voltage)	Controller holds voltage at absorption setpoint. Current tapers as battery fills. Electrochemical reactions complete inside cells.	Current drops below tail current threshold (e.g. < 2% of C-rate) OR time limit reached	Voltage: fixed at absorption Current: tapering toward zero
FLOAT (Main tenance)	Controller reduces voltage to float setpoint. Battery is 100% charged. Small trickle current compensates for self-discharge.	Continuous — controller stays here until load or new bulk cycle needed	Voltage: float setpoint (e.g. 13.6V for LFP) Current: very low (~1% of C)
EQUALIZE (Periodic / Lead-Acid Only)	Applies controlled overcharge to lead-acid batteries to equalise cell voltages and remove sulphation. NOT for lithium batteries.	Fixed time (e.g. 2 hours) as programmed	Voltage: slightly above absorption Never equalise LFP or LiPo — damages cells

The MPPT Advantage — Cold Morning Energy Capture



Figure 3 — MPPT Advantage: Tracks Higher Cold Morning Vmp vs PWM Fixed Voltage

Key setting — temperature compensation: Battery absorption and float voltages change with temperature. LFP: $-3 \text{ mV/}^{\circ}\text{C/cell}$. Lead-acid: $-4 \text{ mV/}^{\circ}\text{C/cell}$. At 0°C , a 12V lead-acid battery needs $\sim 14.8\text{V}$ absorption (not 14.4V). Quality MPPT controllers have a temperature probe input — always connect it to the battery bank for automatic compensation.

Charge Controller Sizing and Programming

Sizing a charge controller correctly prevents damage to the controller and ensures the battery bank charges at the optimal rate. The two critical parameters are: **maximum PV input voltage** (must not exceed controller rating) and **maximum charge current** (must not exceed battery or controller rating).

Sizing Step	Formula	Example (3 kWp, 48V system, LFP)
1. Panel open-circuit voltage (worst case: cold morning)	$V_{oc_max} = N_{series} \times V_{oc_STC} \times \text{temp_correction}$ (temp corr = $1.0 + 0.003 \times \Delta T $ from 25°C)	6 panels series $\times 41.5V \times 1.12$ (at -10°C) = 279V → Select MPPT with $V_{dc_max} \geq 300V$
2. Maximum charge current needed	$I_{charge_max} = P_{array} / V_{battery}$ (at STC)	3 000W / 48V = 62.5A → Select MPPT $\geq 70A$ charge current rating
3. Battery maximum charge current (C-rate limit)	$I_{max_battery} = \text{Capacity_Ah} \times \text{C-rate}$ (LFP: C/2 to 1C; Lead-acid: C/5 to C/10)	200Ah LFP at 0.5C = 100A max charge 62.5A from panels is within limit ✓
4. Verify MPPT input power	$P_{max_input} = V_{mppt_max} \times I_{mppt_max}$ (check controller datasheet)	70A MPPT at 48V battery = 3360W capacity 3000W array within limit ✓
5. Select controller rating	Select next standard size above calculated current Add 25% safety margin	$62.5A \times 1.25 = 78A$ → select 80A MPPT Typical choice: Victron SmartSolar 100/80

Key Charge Controller Settings by Battery Chemistry

Setting	LFP (Lithium Iron Phosphate)	AGM Lead-Acid	Flooded Lead-Acid (FLA)
Absorption voltage (12V system)	14.2–14.6V	14.4–14.7V	14.4–14.8V
Float voltage	13.5–13.8V (or disable float — LFP prefers 50-90% SoC)	13.6–13.8V	13.2–13.5V
Equalize voltage	DISABLE — never equalize LFP	14.8–15.5V (occasionally)	15.0–16.0V (monthly)
Temperature compensation	Minimal: -3mV/°C/cell (or none)	-4mV/°C/cell	-4mV/°C/cell
Tail current (absorption end)	C/20 (e.g. 5A for 100Ah battery)	C/50 (e.g. 2A for 100Ah)	C/100 (e.g. 1A for 100Ah)
Low voltage disconnect	10.5V (12V system) / 21V (24V) / 42V (48V)	11.5V (12V) — protect from deep discharge	11.5V (12V) — more critical than LFP

What comes next — SOLAR-008: DC and AC Systems — the difference between DC-coupled and AC-coupled solar systems, on-grid vs off-grid vs hybrid system architecture, understanding single-phase and three-phase solar installations, and net metering explained.